241401097

PRODUCER CONSUMER PROBLEM SOLUTION USING SEMAPHORE

```
Activities   Terminal                    Mar 24  21:03
                          user123@localhost:~/pc
[user123@localhost ~]$ cd pc
[user123@localhost pc]$ ls
a.out  pc  producer.c
[user123@localhost pc]$ vi producer.c
[user123@localhost pc]$ cat producer.c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

sem_t empty;
sem_t full;
sem_t mutex;

void *producer(void *arg)
{
        int item;

        for(int i=0; i<BUFFER_SIZE; i++){

                item = i;
                sem_wait(&empty);
                sem_wait(&mutex);

                buffer[in] = item;
                printf("Producer: %d\n",item);
                in = (in + 1)%BUFFER_SIZE;

                sem_post(&mutex);
```

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```
                sem_post(&mutex);
                sem_post(&full);
        }
}

void *consumer(void *arg)
{
        int item;

        for(int i=0; i<BUFFER_SIZE; i++) {
                sem_wait(&full);
                sem_wait(&mutex);

                item = buffer[out];
                printf("Consumed: %d\n",item);
                out = (out + 1) % BUFFER_SIZE;

                sem_post(&mutex);
                sem_post(&empty);
        }
}

int main()
{
        pthread_t p, c;

        sem_init(&empty, 0, BUFFER_SIZE);
        sem_init(&full, 0, 0);
        sem_init(&mutex, 0, 1);

        pthread_create(&p, NULL, producer, NULL);
        pthread_create(&c, NULL, consumer, NULL);
```

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```
int main()
{
        pthread_t p, c;

        sem_init(&empty, 0, BUFFER_SIZE);
        sem_init(&full, 0, 0);
        sem_init(&mutex, 0, 1);

        pthread_create(&p, NULL, producer, NULL);
        pthread_create(&c, NULL, consumer, NULL);

        pthread_join(p, NULL);
        pthread_join(c, NULL);

        sem_destroy(&empty);
        sem_destroy(&full);
        sem_destroy(&mutex);

        return 0;
}
[user123@localhost pc]$ gcc producer.c
[user123@localhost pc]$ ./a.out
Producer: 0
Producer: 1
Producer: 2
Producer: 3
Producer: 4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[user123@localhost pc]$
```

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